A Comprehensive Survey of Infant Cry Classification Research trends and methods: A Systematic Review

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***Abstract*— Infant cry classification has become a crucial focus in pediatric healthcare due to its potential in early detection of health issues and understanding infant needs. This systematic review aims to analyze the current trends and methods in infant cry classification research, particularly focusing on the evolution from traditional techniques to advanced deep learning models like Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN). A total of 35 studies were examined, covering various methods such as machine learning and deep learning, with most utilizing acoustic features for cry analysis. The results indicate that CNN-RNN models outperform traditional methods in accuracy, though challenges remain, such as the lack of standardized datasets and the difficulty in integrating multi-modal data like acoustics, body language, and neurophysiological signals. The review highlights the need for future research to focus on building more diverse datasets and improving multi-modal data integration to enhance the effectiveness of infant cry classification models for real-world applications.**

Keywords—infant cry, baby cry, infant cry classification, baby cry recognition, infant cry detection

I. INTRODUCTION

The recognition and interpretation of infant cries has gained considerable attention in recent years because of its potential to detect health issues early and assist in understanding infant needs and emotions [1] [2]. Recognizing and interpreting infant cries presents a significant challenge in pediatric care, especially during the early stages of life. Crying is typically the first indication of discomfort or health issues in infants, but differentiating between typical crying and cries that suggest a serious condition can be highly challenging [3][4]. The objective of this article is to provide a thorough examination of the research trends and methodologies in infant cry classification. By systematically reviewing existing literature, the study aims to identify the most effective techniques and approaches used in the field, highlight current advancements, and uncover gaps in the research [5][6][7][8][9]. This comprehensive survey seeks to offer insights that will guide future studies, ultimately contributing to the improvement of infant healthcare through enhanced cry recognition and analysis.

A. Related Survey Paper

In the context of infant cry classification, a survey paper serves as an important instrument for summarizing various research and methods developed in understanding and classifying infant cries[10] [11], [3]. This survey paper will explore various aspects, ranging from signal analysis techniques to the utilisation of modern technologies such as machine learning and deep learning to identify the configuration and interpretation of infant vocalisations [12], [13], [14], [15].

- Vankudre G et al. conducted a study that aimed to explain how video clips can be used to recognise infant emotions, here we can explore the baby's cry from the baby's emotional side [1]. There is still room for improvement in the use and development of more advanced classification algorithms, including neural network combinations such as RNN-CNN, to improve emotion recognition accuracy.
- Alishamol et al. explained in their research that to overcome the problem of busy parents not being able to monitor and meet their babies' needs immediately, they designed a system to identify babies' emotions based on the analysis of their crying signals, categorising them into 'hungry', 'sleeping', and 'uncomfortable', we can develop methods on the proposed algorithm[16]. This study may not have considered variations in infant crying that may occur among different cultures or ethnicities.
- Nadia et al. classified five types of emotions or needs expressed, namely appetite, drowsiness, unease, stomach discomfort, and cues suggesting the baby needs to burp, from this research we can explore the extraction of baby crying sounds[17]. In this article Applying CNN-RNN models in real-time applications can be challenging due to computational complexity and resource requirements.
- Laguna et al. analysis of infant cries with neurophysiological and behavioural signals, showing that each state (such as hunger, sleepiness, fussiness, burping, and distress) is characterised by different

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acoustic, neurophysiological, and behavioural patterns. In the study, the sample given was still small [18]. The gap in this research is from effectively integrating and synchronising different types of data to provide a thorough understanding of infant crying.

- Turgut, investigated the effectiveness of various methods, including hand-crafted features, spectral images, and artificial neural network models, in classifying infant cries [19]. The gaps in this research can increase the effectiveness and application of the CNN-RNN model in the classification of baby cries.
- Zhang et al. The proposed method includes an improved Dempster-Shafer evidence theory (DST) based on Wasserstein distance and Deng entropy, effectively minimising conflicts among deep learning models and improving the accuracy of infant cry recognition [20]. A gap could exist in applying the model across diverse datasets or real-world scenarios.

By analyzing research trends, the study highlights the evolution of methods and approaches in infant cry classification. This insight can guide future research directions, helping to identify which areas are gaining traction and which may be underexplored.

II. METHODOLOGY

This part outlines the approach utilized for conducting the investigation. It also covers the research inquiries, the methodology for identifying pertinent studies, the criteria for inclusion/exclusion, the standards for evaluation, and the study's scope. Additionally, it provides a concise overview of existing relevant review studies [21] [22]. The planning stage is depicted in Figure 1.

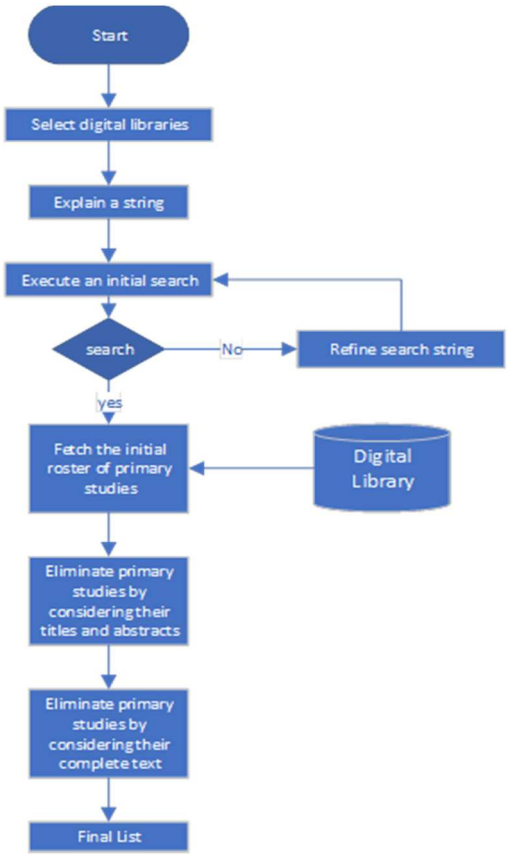


Fig. 1. Literature Search and Selection

Figure 1 describes literature search and selection for a survey paper involve a systematic process of identifying and choosing relevant studies to comprehensively address the research topic. This procedure usually commences by formulating the research inquiries and setting up criteria for inclusion and exclusion. Subsequently, various databases and digital libraries are searched using predefined search terms and strategies. The primary search outcomes undergo screening based on titles and abstracts to pinpoint potentially pertinent studies. Subsequently, a thorough examination of full texts is carried out to narrow down the selection, eliminating studies that fail to meet the established criteria.

A. Research Questions

The research question was constructed using the criteria of PICOC (Population, Intervention, Comparison, Outcome, Context). Table I shows the formulation of the research questions built from PICOC.

TABLE I. PICOC-BASED RESEARCH QUESTION FORMULA

PICOC	Question Formula
Population	Infant, baby
Intervention	detection, classification, model, methode, dataset
Comparison	Comparison of models, methods and algorithms in different datasets
Outcomes	Average accuracy of infant cry classification and the effectiveness of infant cry classification method
Context	hospital and academic studies, industry, small and large datasets

Table I Using the PICOC framework, researchers can formulate clear research questions, determine appropriate inclusion and exclusion criteria, and effectively guide the literature search process to systematise the literature review on infant cry classification.

Derived from the content presented in Table I, the research queries employed in the systematic literature review are delineated. The research questions utilized in this systematic literature review are outlined in Table II.

TABLE II. RESEARCH QUESTION

	Research Questions
RQ1	What are the most published journals on infant cry classification?
RQ2	What is the dataset used in the research?
RQ3	What are the methods used in the research?
RQ4	What improvements have researchers made?

B. Search Strategies

In the phase depicted in Figure 1, the process involves selecting a digital library, constructing search strings, executing searches, and procuring literature that corresponds to the search criteria. Prior to conducting searches, the digital library selection process must be completed. It is imperative to choose appropriate digital libraries or databases to enhance the likelihood of discovering pertinent literature [23], [22], [24], [25]. The digital bibliographies that will be used in this systematic literature review are Science Direct, IEEE Access, ACM Digital Library, Springer.

After determining the digital library, the next step is to designing the search string, as for the search string used are: (infant OR baby) AND cry AND (detect* OR classificat*).

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The search process was carried out by retrieving articles from the digital library mentioned above totalling 2369.

C. Selection Criteria

After obtaining the reference list, the next step was to select each piece of literature to determine whether it should be included for data extraction and analysis. To make the selection, will be used inclusion and exclusion criteria:

a) Inclusion, The inclusion criteria to consider in this study was the selection of articles that fit the purpose of the study and the topic under study. Articles should address techniques, methods or approaches in categorising infant vocalisations. In the analysis and classification of infant cries, the methods or approaches used should be clear, such as the use of digital signal processing techniques, the development of classification models using artificial intelligence, or other approaches. In addition, the article should meet adequate research quality standards, including appropriate research design, use of representative samples, accurate data analysis, and in-depth interpretation of results. Finally, the selected articles should have publication dates that are relevant to recent developments in the field of infant cry classification, while still considering classic articles that made important contributions to the understanding of the topic.

b) Exsclusion, Articles that were not relevant to infant crying classification, did not use appropriate methods, did not meet research quality standards, and were not published in peer-reviewed scientific journals or conference proceedings, as well as articles that were duplicates of previous publications or had been previously analysed in the same literature review, fell under the exclusion criteria. Articles that lacked strong validation, such as application experiments for infant crying, were also excluded.

The selection stage will be carried out in several stages, namely selection of titles, selection of abstracts and keywords, and selection of the content of the article.

After selecting article titles, the 2369 articles were reduced to 200 articles taken from the top 50 articles from each digital library. Then the abstract and keyword selection was carried out on the article, resulting in 99 articles. The last stage in the selection was content selection which resulted in 35 articles, where the articles were taken according to the relevant topics and using the method with the best accuracy rate.

III. Results And Discussion

A. RQ1: What are the most published journals on infant cry classification?

The most active journals in publishing research on the classification of baby cries, which are taken as relevant to the related criteria are in IEEE Access as many as 19. Figure 2 provides an explanation of the number of journals used for this article.

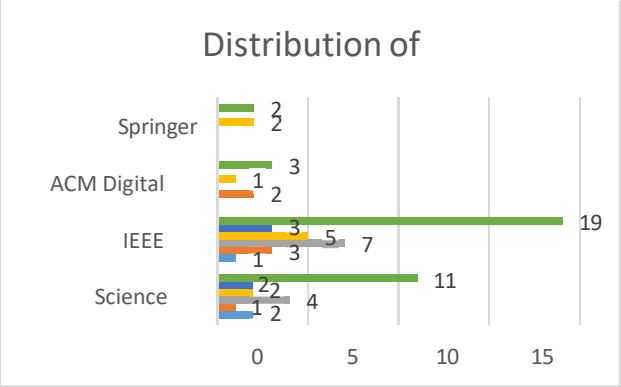


Fig. 2. The most published journals on infant cry classification

B. RQ2: What is the dataset used in the research?

The dataset employed in this study is a collection of recorded infant cries collected from various sources, including clinics, laboratories, or in-person surveys. Using these datasets, researchers can develop and test classification models of infant cries that can aid in the early diagnosis of health disorders or provide insights into the emotional state of infants. Figure 3 explains what datasets have been used from this article. It can be seen that most of the existing datasets belong to other people because the datasets are taken without providing information on where the baby cries are taken from. In addition, most of the datasets are also taken from hospitals.

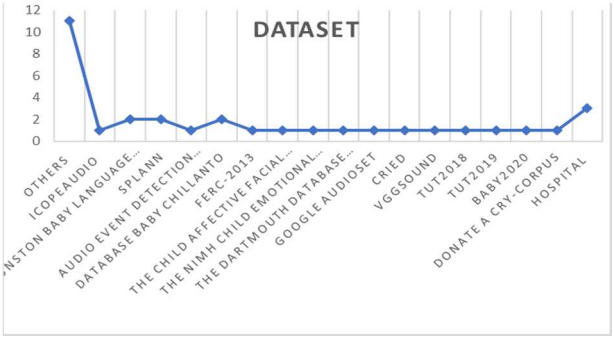


Fig. 3. Datasets used in this literature review article

In the study that encompasses the diverse range of variations in infant cries, particularly focusing on samples indicating breathlessness, utilizing a specific database (Baby Chillanto Database) for evaluation of the proposed method [30], [31]. While some articles used datasets taken from recording infants in hospitals, here we grouped them into other categories [32], [33], [34], [35], [36], [37].

C. RQ3: What are the methods used in the research?

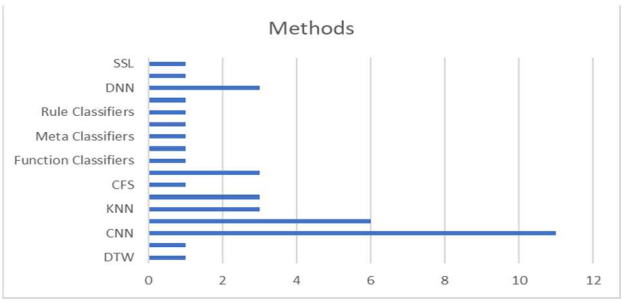


Fig. 4. Methods used in the articles in the literature review

Out of the 35 literatures used, there are various methods used to build models to classify infant cries. From the 35

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literatures used, there are various methods used to build models to classify baby cries. two methods used are machine learning and deep learning [37], [38], [39], [40], [41], [42],

[43], [44], [45], [46], [47], [48], [49]where some methods used are Support Vector Machine (SVM), KNN, Random Forest and others, while methods used in deep learning such as ANN and CNN are shown in Figure 4.

D. What improvements have researchers made?

Researchers have achieved significant advancements in the classification of infant cries, leveraging technology to enhance early detection and intervention in pediatric healthcare [50][51][52][53][2][48].

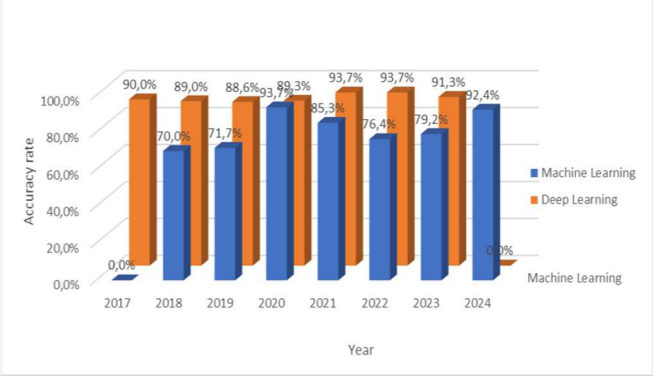


Fig. 5. A line chart showing the comparison of the use of methods with the results of the level of accuracy regarding infant cry in 2017 – 2024.

Figure 5 presents a comparison of the methodologies used in studies concerning the classification of infant cries. The accuracy of the methods is represented on the horizontal axis, where methods used in ML such as Support SVM, KNN, Random Forest and others, while methods used in deep learning such as ANN and CNN [8],[49],[50],[51]. The vertical axis shows the year when the method was applied. Each data point on the graph corresponds to a particular method used in the infant cry classification research, which displays its level of accuracy [52],[53],[47],[54]. This graphical representation provides an insight into the development of the methodologies used in this field, showing patterns or progress in accuracy rates from year to year. By analysing this visual representation, researchers can assess the effectiveness of various methods over time, thus helping in identifying promising approaches and areas that can be further improved in infant cry classification research [55], [56].

IV. Conclusion

Based on the systematic review, it is evident that deep learning methods, particularly Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), are among the most effective approaches for classifying infant cries due to their ability to capture intricate patterns in acoustic signals. These models demonstrate superior accuracy when compared to traditional machine learning techniques such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN). However, several research gaps persist, including the lack of standardized and diverse datasets, which limits the generalization of models across different populations and cry types. Additionally, the integration of multi-modal data, such as combining acoustic signals with body language and neurophysiological cues, remains underexplored, and more advanced methods are needed to synchronize these data effectively. Future research should focus on addressing these gaps by developing larger, more diverse datasets and enhancing the fusion of multi-modal information to improve the accuracy and applicability of infant cry classification models

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